

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T33 · Graphs of Functions

Questions

Student book · questions with writing space

■ 22 pages

FAMILY MAP

Graphs of Functions

F01	Generate a table and draw a nonlinear graph	8 Q 25 marks
F02	Recognise a function graph family	4 Q 9 marks
F03	Sketch key features without a full table	2 Q 7 marks
F04	Solve using graph roots or intersections	5 Q 17 marks

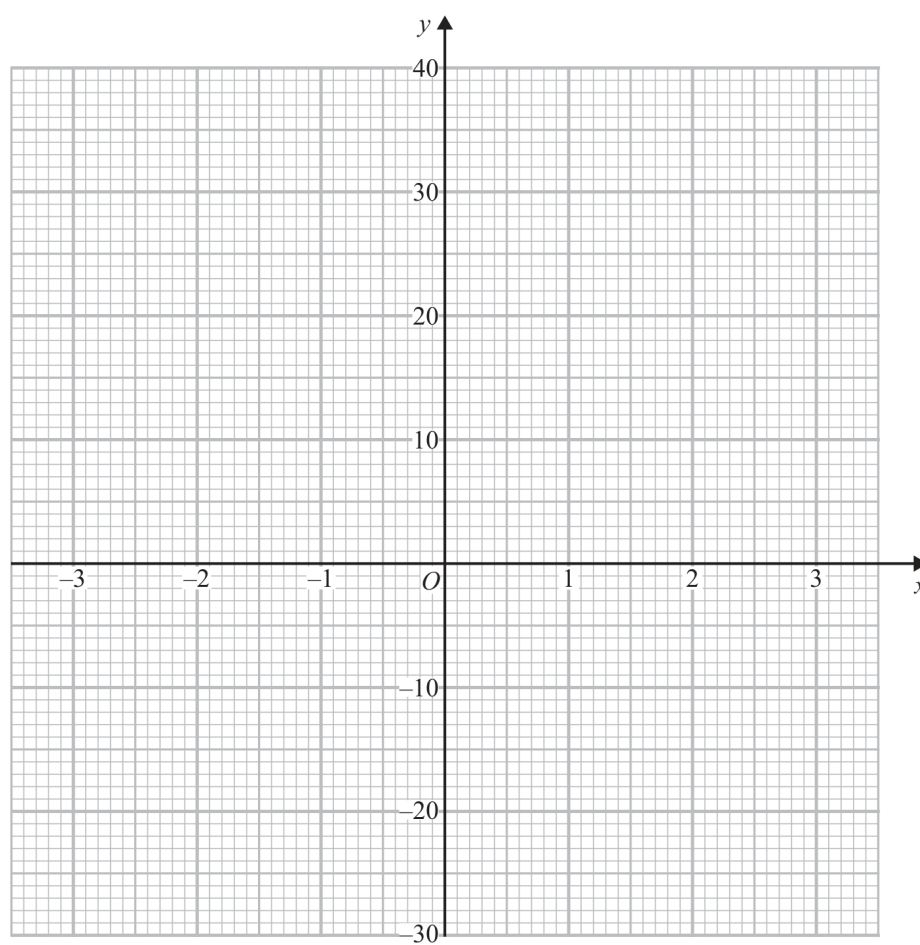
Working Unit 07

4MA1-1H Q10

UNIT 1**3 marks**10 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	-3	-2	-1	0	1	2	3
y		-9	0	3	6		36

(1)

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leq x \leq 3$ 

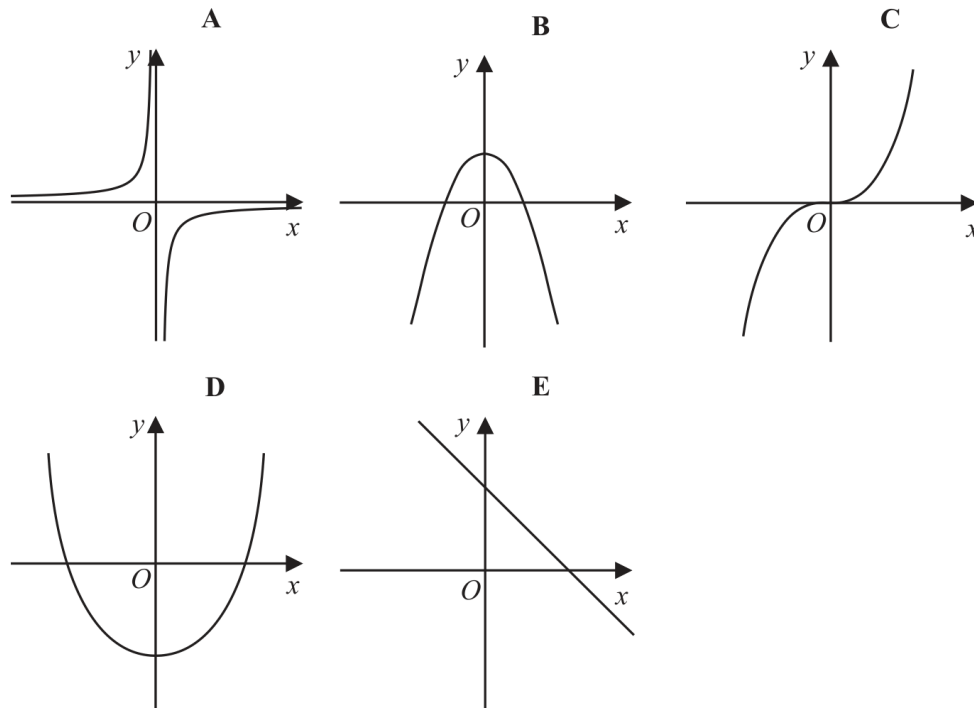
(2)

Working Unit 20

4MA1-2H Q14

UNIT 1
2 marks

14 Here are five graphs.

(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$
(1)(b) Write down the letter of the graph that could have the equation $y = 2x^3$
(1)

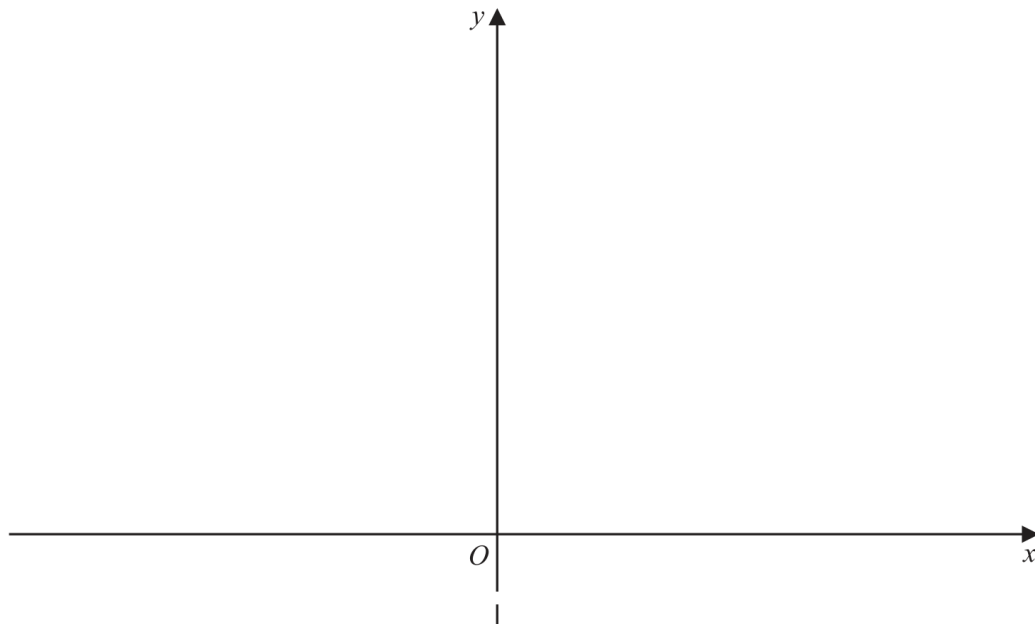
Working Unit 28

4MA1-2H Q25

UNIT 1**3 marks**

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



(3)

Question 11

4MA1-1H Q12 (a), (b)

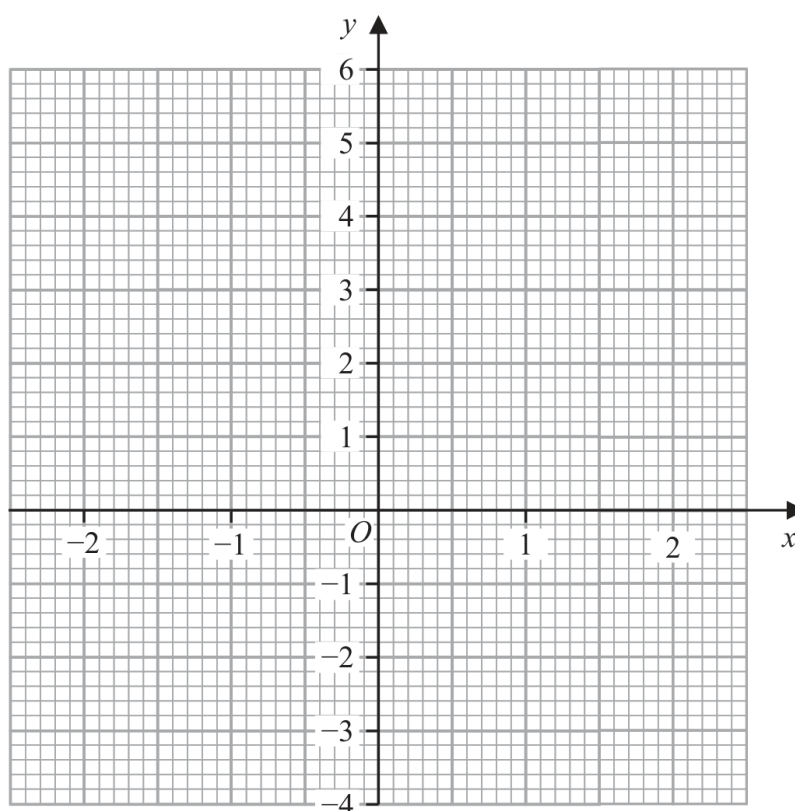
MODULAR UNIT 1

4 marks

12 (a) Complete the table of values for $y = x^3 - 3x + 1$

x	-2	-1	0	1	2
y				-1	

(2)

(b) On the grid, draw the graph of $y = x^3 - 3x + 1$ for values of x from -2 to 2

4MA1-1H Q12 (c)

1 marks

The figure displays four separate Cartesian coordinate systems, each with a horizontal x -axis and a vertical y -axis. The origin of each system is labeled with the letter O .

- Graph A:** Shows a function with two branches. One branch is in the first quadrant, starting from a high y -value near the y -axis and decreasing towards the x -axis. The other branch is in the third quadrant, starting from a low y -value near the y -axis and increasing towards the x -axis.
- Graph B:** Shows a function with two branches. Both branches are in the upper half-plane ($y > 0$). One branch is in the first quadrant, starting from a high y -value near the y -axis and decreasing towards the x -axis. The other branch is in the second quadrant, starting from a high y -value near the y -axis and decreasing towards the x -axis.
- Graph C:** Shows a function with two branches. One branch is in the second quadrant, starting from a high y -value near the y -axis and decreasing towards the x -axis. The other branch is in the fourth quadrant, starting from a low y -value near the y -axis and increasing towards the x -axis.
- Graph D:** Shows a function with two branches. Both branches are in the lower half-plane ($y < 0$). One branch is in the third quadrant, starting from a low y -value near the y -axis and increasing towards the x -axis. The other branch is in the fourth quadrant, starting from a low y -value near the y -axis and increasing towards the x -axis.

Question 15

4MA1-2H Q4

MODULAR UNIT 1

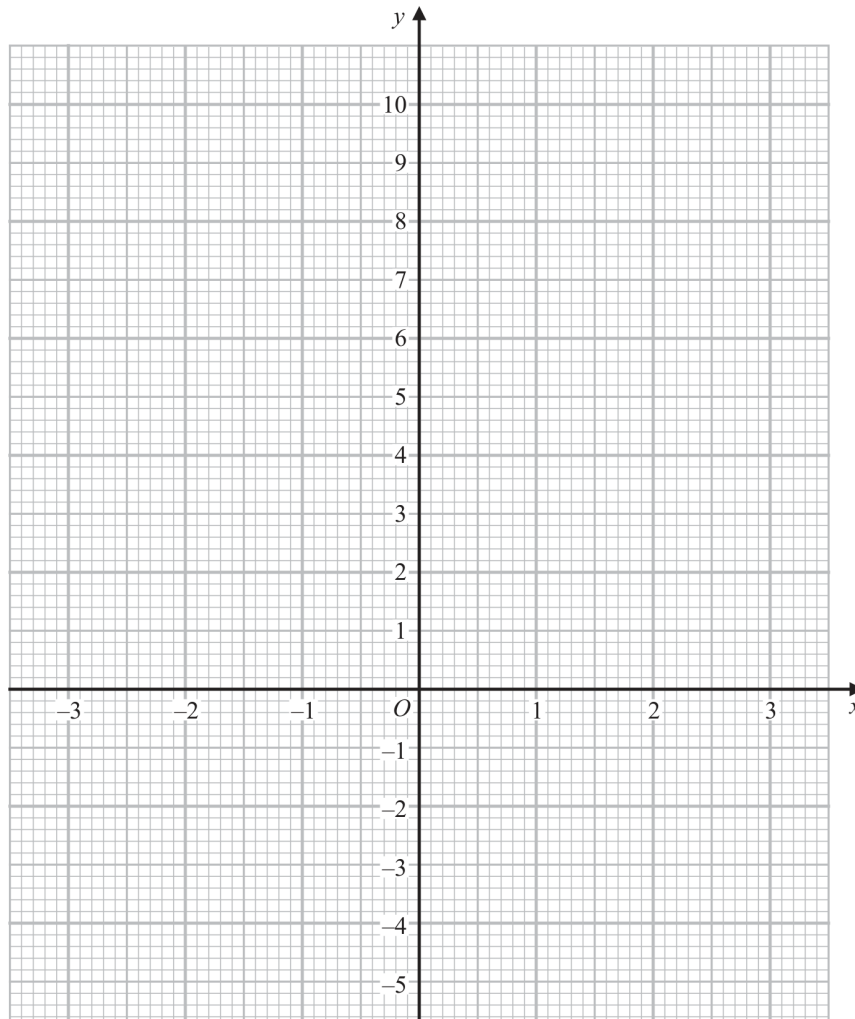
4 marks

- 4 (a) Complete the table of values for
- $y = x^2 - x - 4$

x	-3	-2	-1	0	1	2	3
y		2			-4		

(2)

- (b) On the grid below, draw the graph of
- $y = x^2 - x - 4$
- for values of
- x
- from -3 to 3



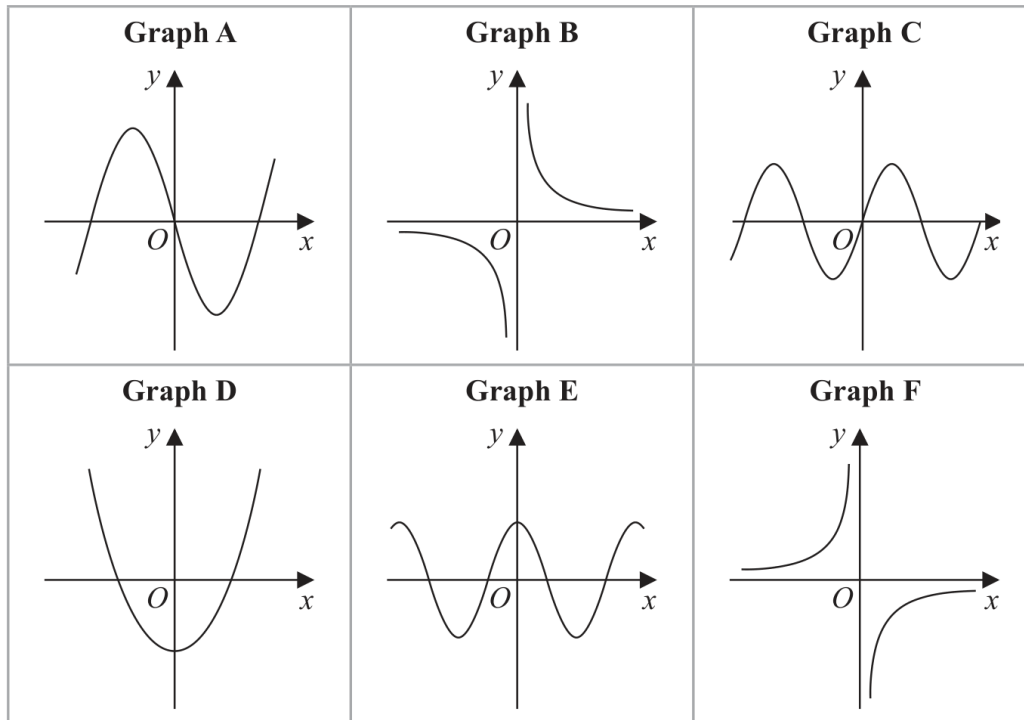
Question 23

4MA1-2H Q18

MODULAR UNIT 1

3 marks

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	
$y = -\frac{3}{x}$	
$y = 4x^3 - 5x$	

Question 08

4MA1-1H Q11 M01, M02

MODULAR UNIT 1

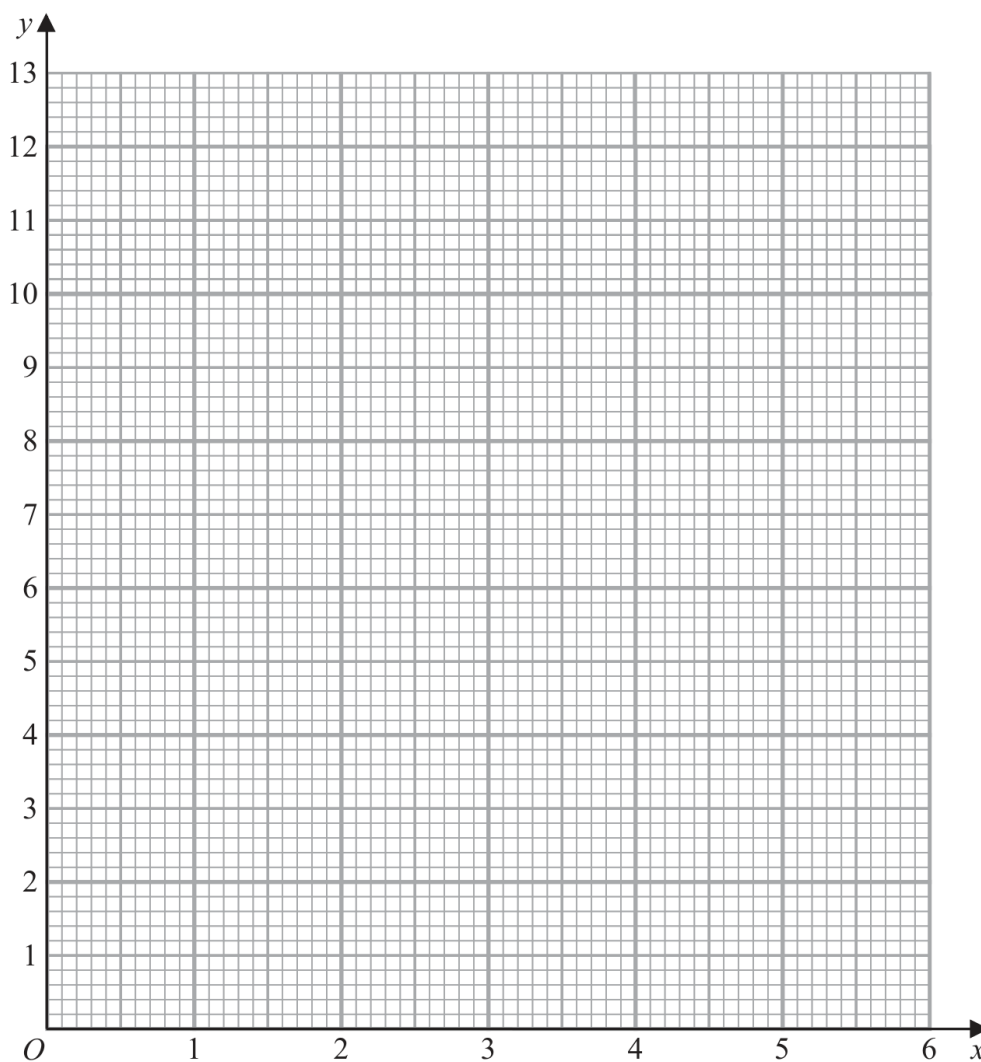
4 marks

- 11 (a) Complete the table of values for $y = 2\left(x + \frac{1}{x}\right)$

x	0.5	1	2	3	4	5	6
y			5	6.7			12.3

(2)

- (b) On the grid, draw the graph of $y = 2\left(x + \frac{1}{x}\right)$ for $0.5 \leq x \leq 6$



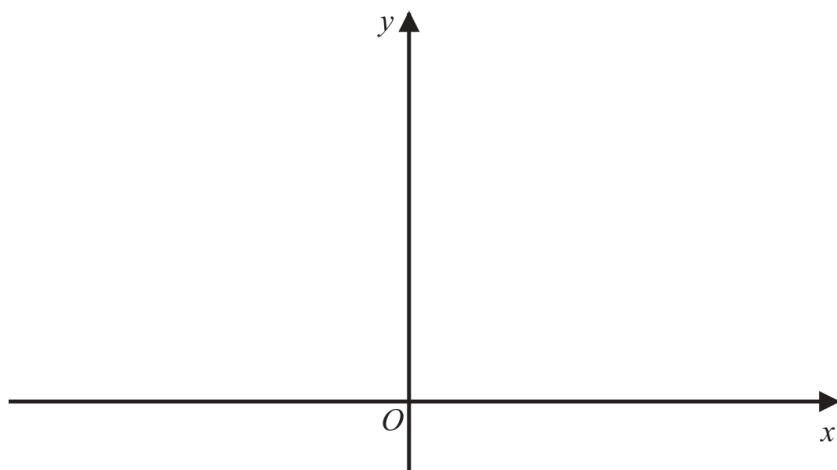
20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.

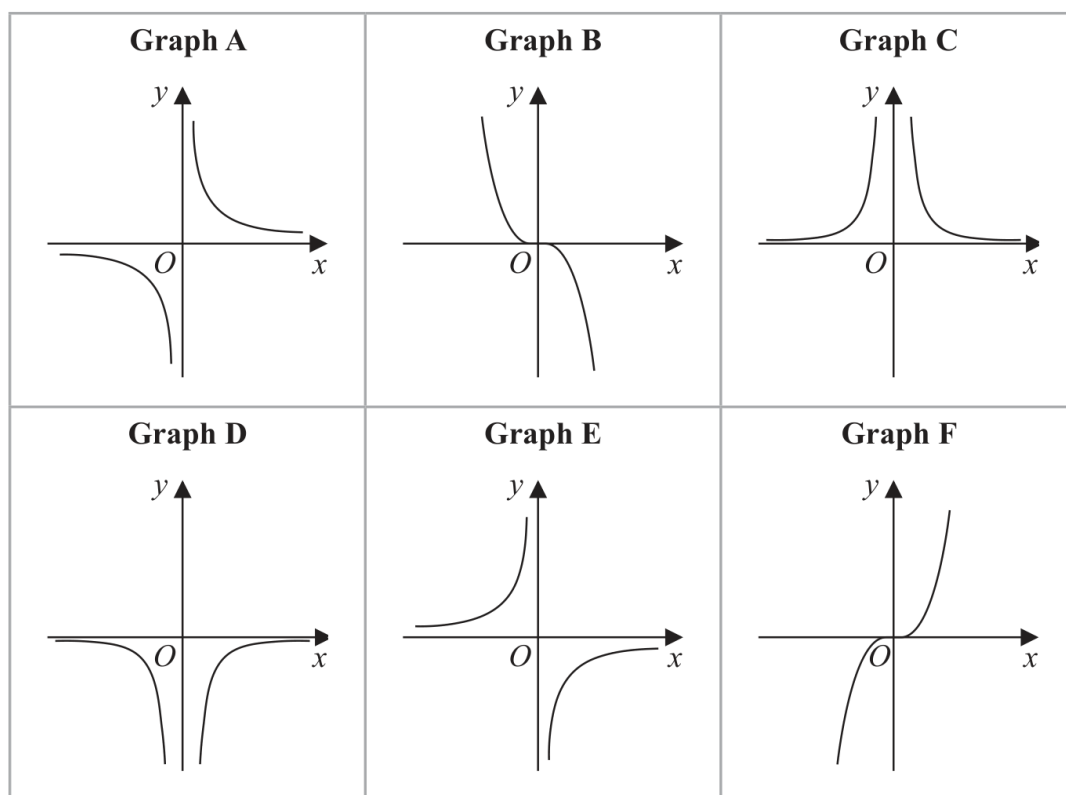
On your sketch show clearly, in terms of a ,

(i) the coordinates of any points of intersection of **C** with the coordinate axes,

(ii) the coordinates of the turning point.



15 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

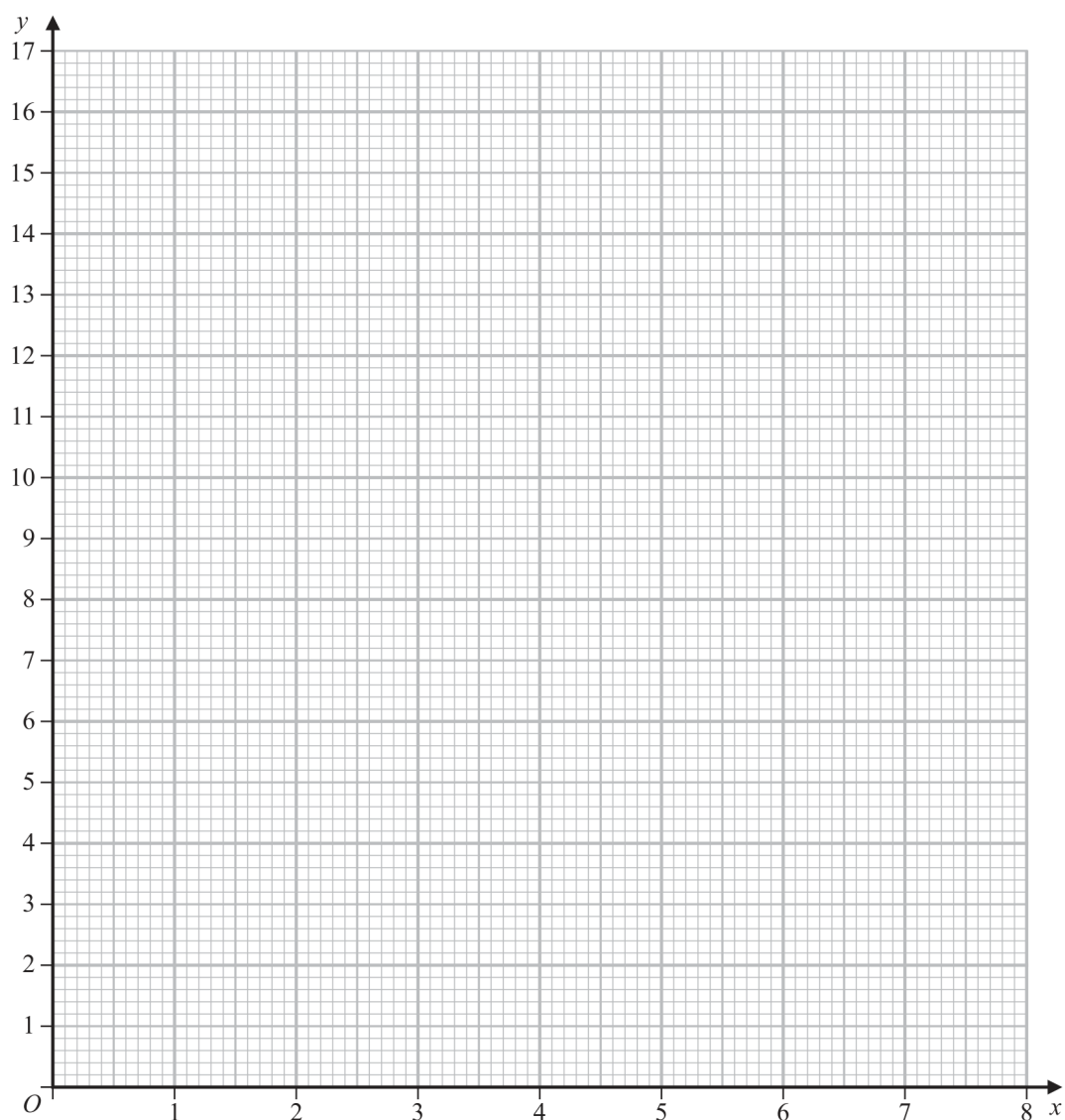
Equation	Graph
$y = \frac{2}{x^2}$	
$y = -\frac{1}{2}x^3$	
$y = -\frac{5}{x}$	

15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leq x \leq 8$

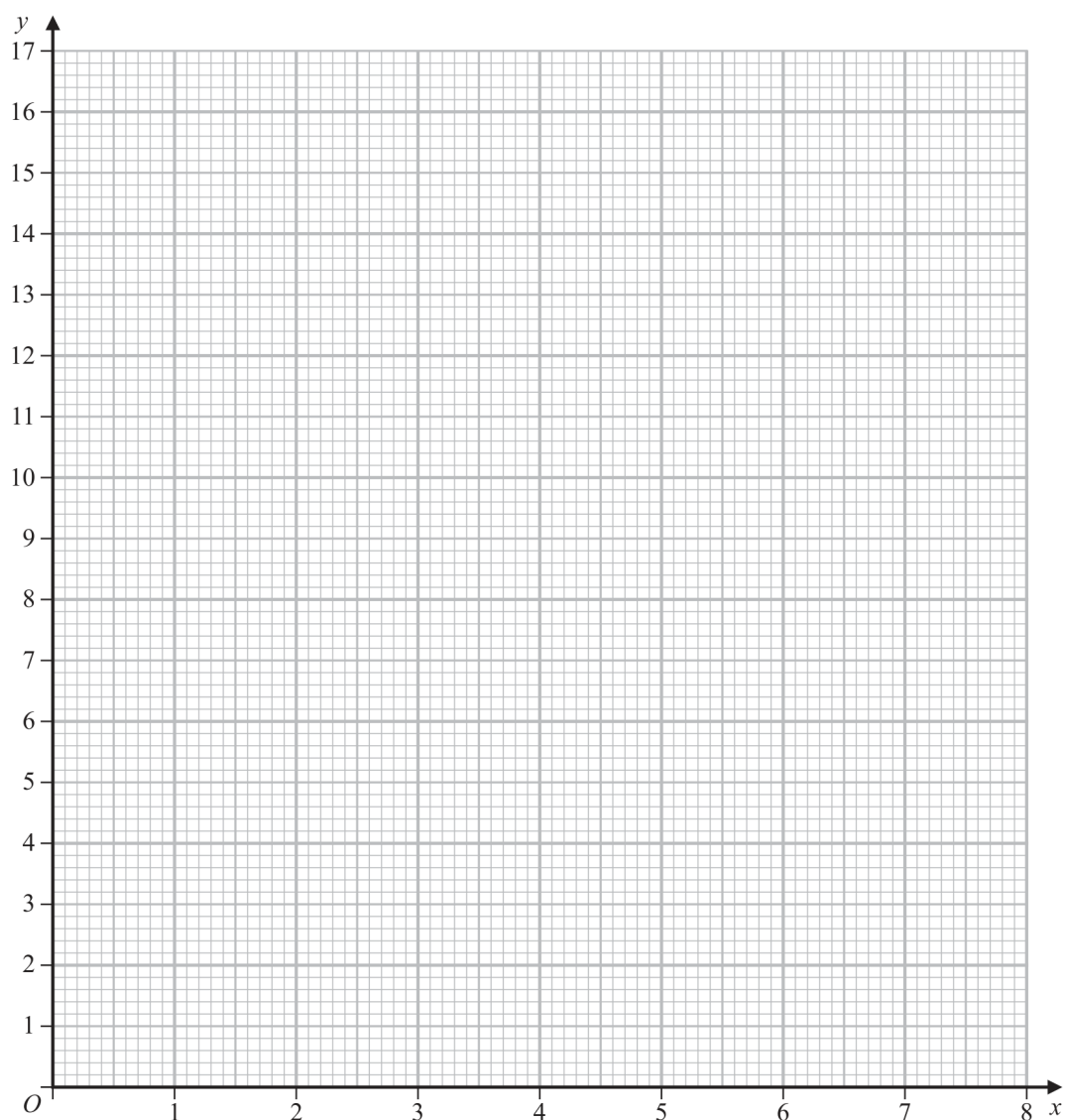


15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leq x \leq 8$

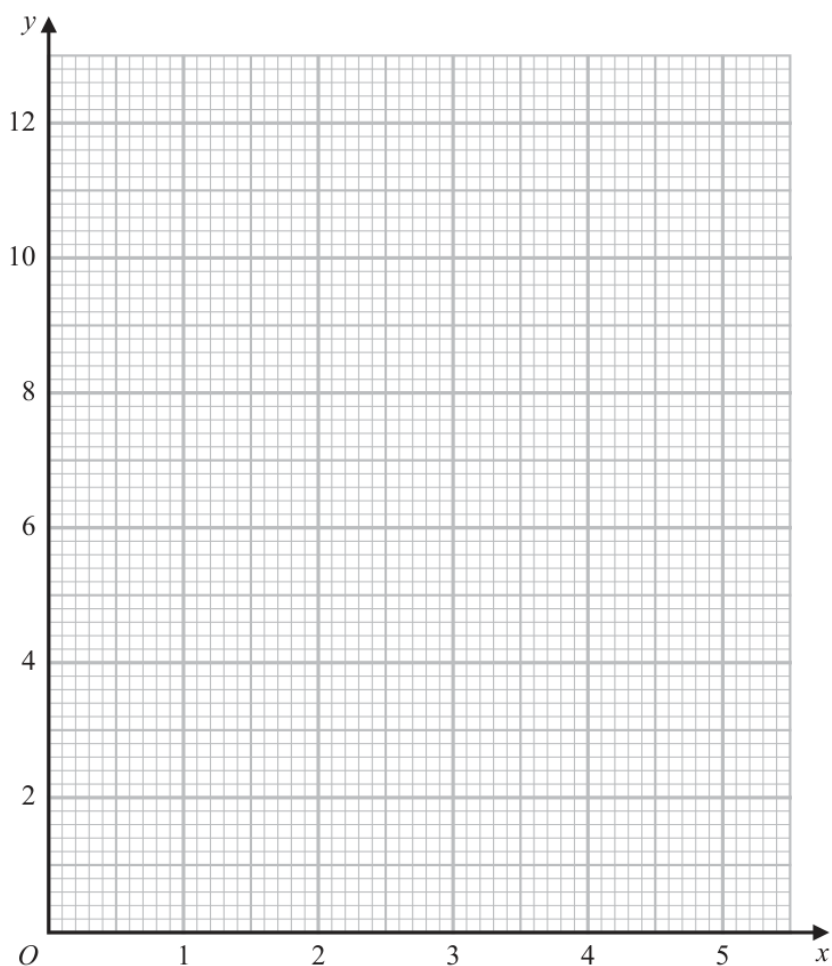


14 (a) Complete the table of values for $y = \frac{2}{x} \left(5 - \frac{1}{x} \right)$

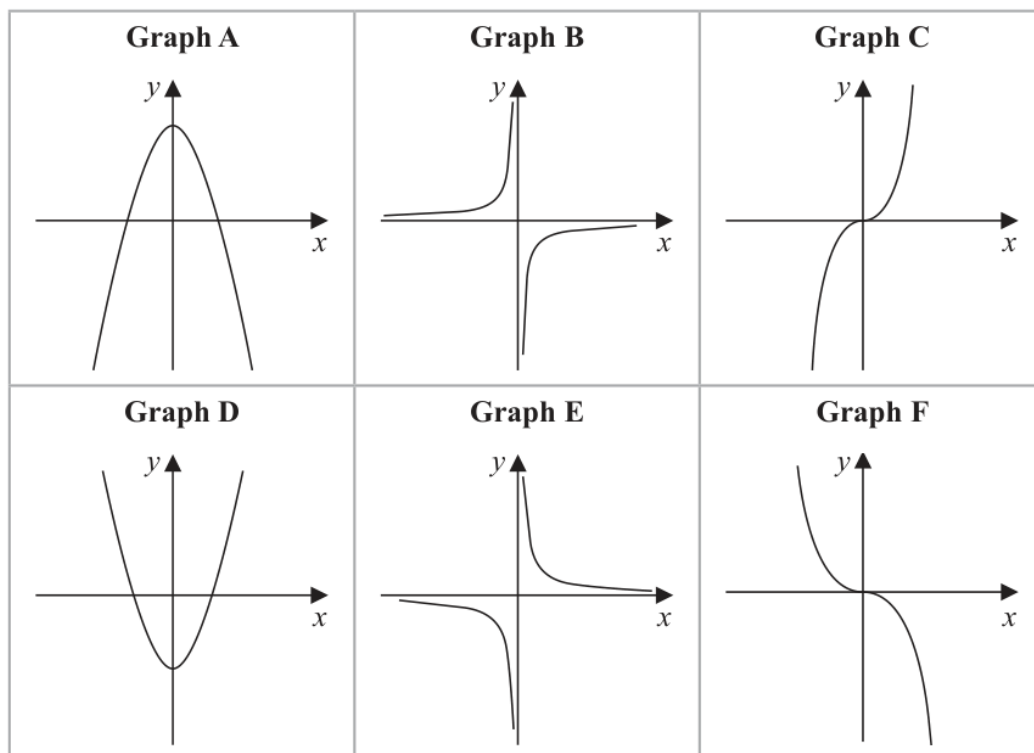
x	0.5	1	2	3	4	5
y		8		3.1	2.4	1.9

(1)

(b) On the grid, draw the graph of $y = \frac{2}{x} \left(5 - \frac{1}{x} \right)$ for $0.5 \leq x \leq 5$



11 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = -\frac{2}{x}$	
$y = 5 - x^2$	
$y = -2x^3$	

4MA1-1H Q23 M01

5 marks

- Show clear algebraic working.

Question 15 (continued)

4MA1-1H Q23 M01

MODULAR UNIT 2

5 marks

$k =$

Lined area for student response.

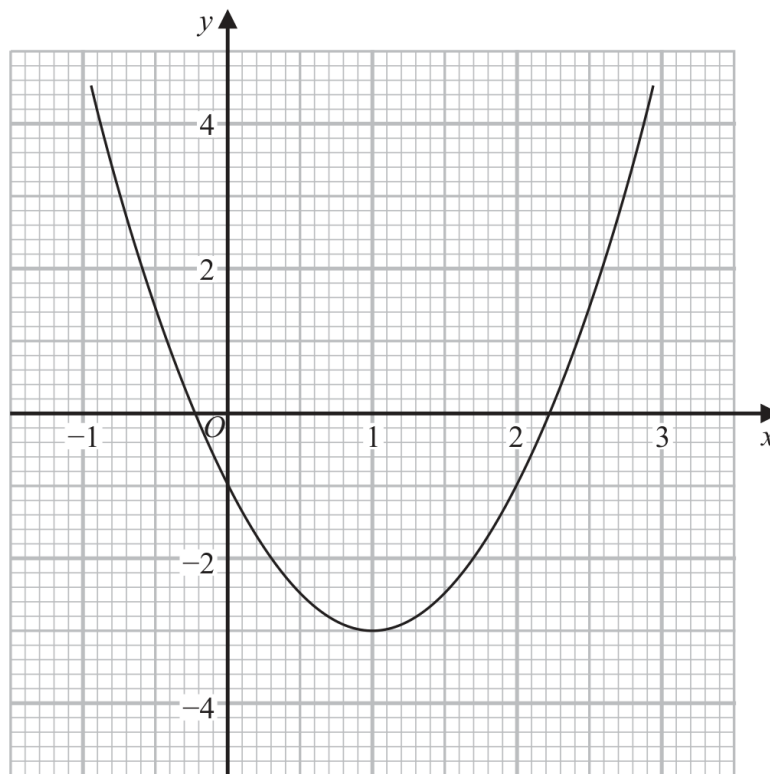
Question 32

4MA1-2H Q21 Q21(a)

MODULAR UNIT 2

2 marks

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$.
Give your solutions correct to one decimal place.

Question 33

4MA1-2H Q21 Q21(b)

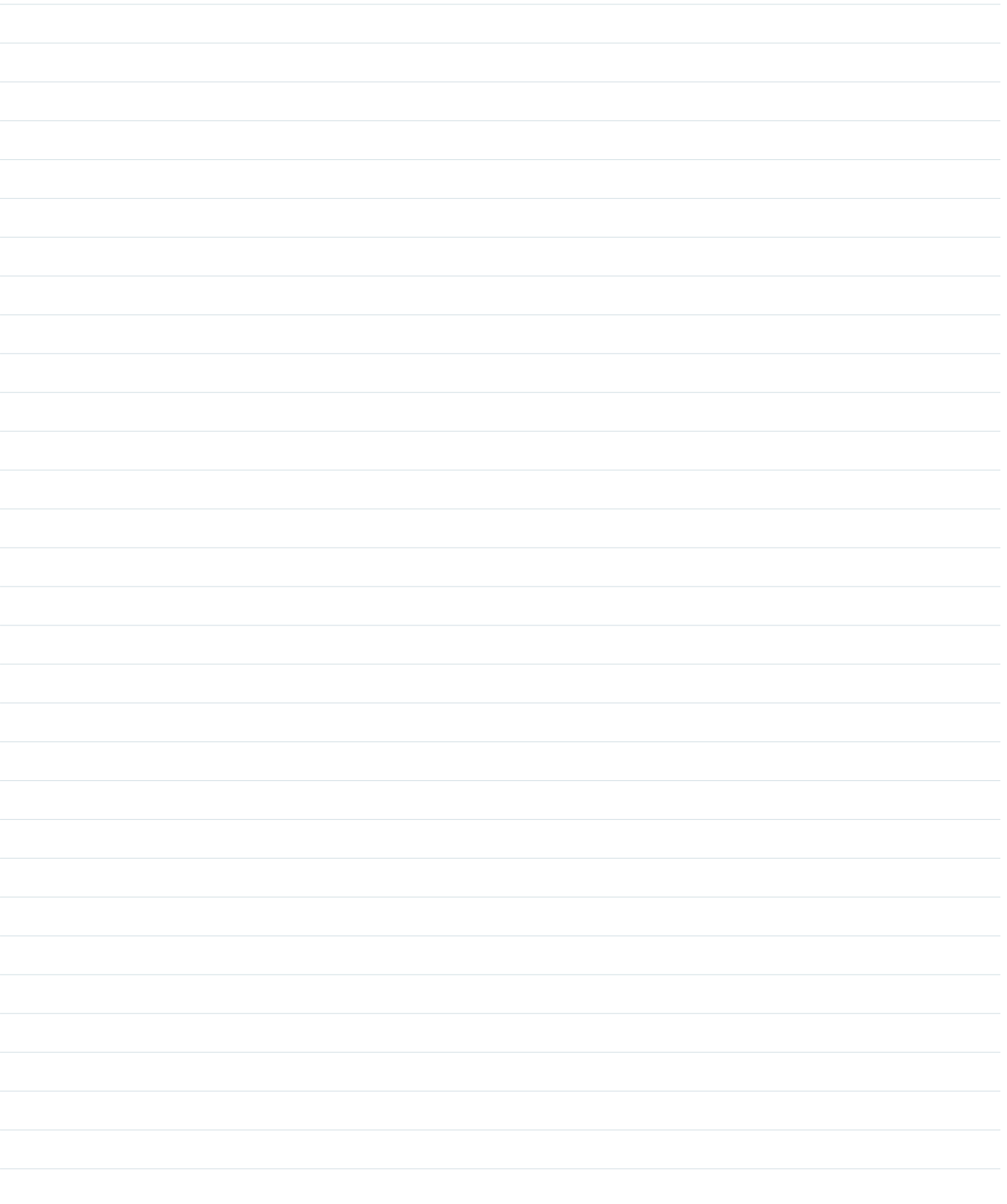
MODULAR UNIT 2

3 marks

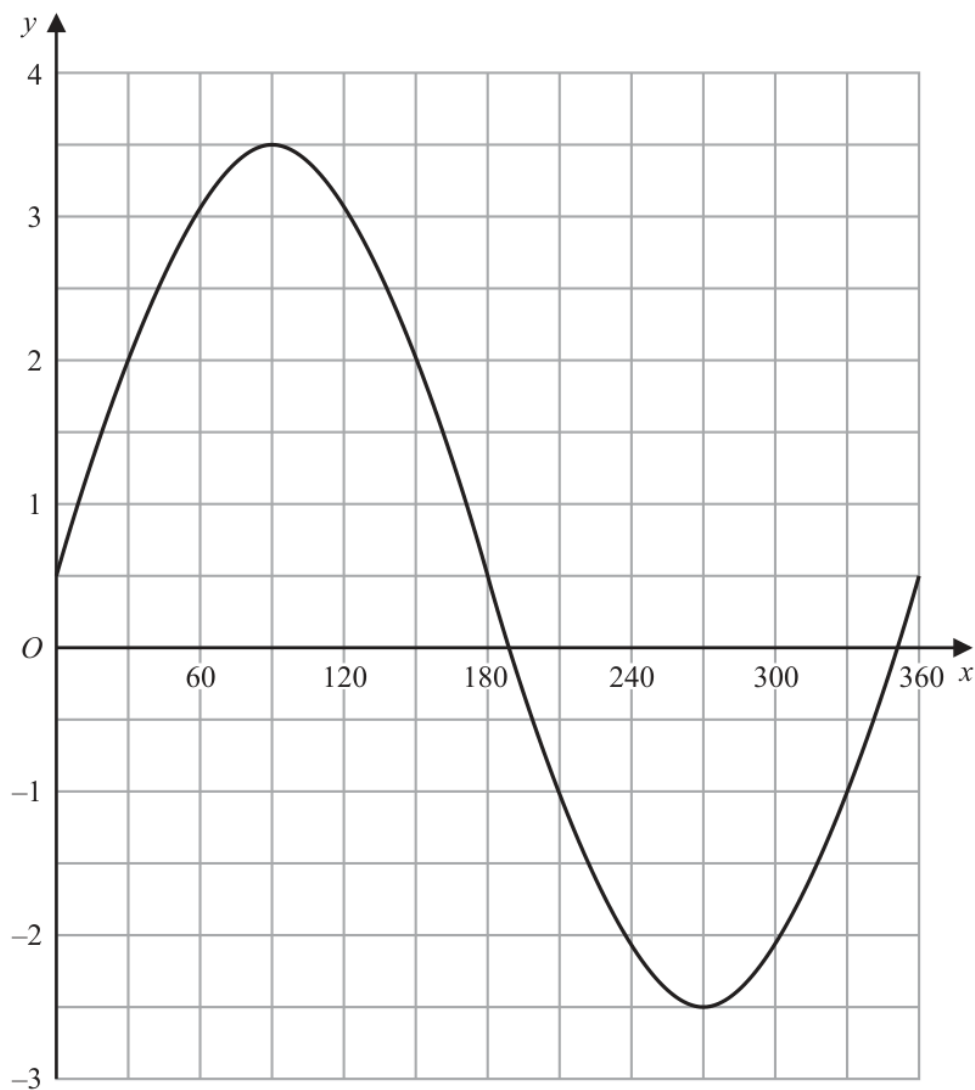
- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$

Show your working clearly.

Give your solutions correct to one decimal place.



24 The graph of $y = a \sin x^\circ + b$ is drawn on the grid.



Find the value of a and the value of b

$a =$

$b =$

(Total for Question 24 is 2 marks)

(1)

x	0.5	1	2	3	4	5	6
y	4.25			2.17	2.5	2.9	3.33

A blank Cartesian coordinate system with x and y axes ranging from 0 to 6. The grid is composed of major lines every 1 unit and minor lines every 0.2 units. The origin is labeled O.

- (c) By drawing a suitable line on the grid, find estimates for the solutions of the equation $x + \frac{4}{x} = 6$

Give your answers correct to one decimal place.

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(2)

(Total for Question 12 is 5 marks)